



EAST WEST UNIVERSITY

CSE475: Machine Learning
[Fall 2025]

Related Work

Course Code : CSE475
Course Title : Machine Learning
Section : 02
Group Number : 02

Submitted by:

Student Name	Student ID
Rafia Noor Nithin	2022-2-60-120
Fahimah Karim Oishi	2022-2-60-134
Mohammad Ashiquzzaman Hadi	2021-3-60-143
Sahla Fabia Safa	2021-2-60-061

Related Work

Ref.	Title	Dataset Description	Methods	Results	Pros	Cons	Future Work
[1]	Image-Based Air Quality Analysis Using Deep Convolutional Neural Network	Image-PM _{2.5} (Beijing) — 591 images ($\geq 500 \times 500$) matched with hourly U.S. Embassy PM _{2.5} . Classes: Good < 75, Moderate 75–115, Severe > 115. Dataset: http://www.tour-beijing.com/real_time_weather_photo/	Transfer learning (MatConvNet, ImageNet). Fine-tuned CNN with 3-class softmax & dropout; also 4096-D CNN features → Random Forest (500 trees). Data $\times 4$ by splitting + flipping.	CNN 68.74 % accuracy; RF 63.62%. Qualitative results and prior comparison included	1. End-to-end vision (no explicit transmission / depth). 2. Public dataset + clear setup. 3. Two TL strategies compared.	1. 3-class coarse labeling; small N → ~69% acc. 2. Single-city domain.	1. Collect more images across cities/seasons 2. try other CNNs (DAG/RCNN) 3. consider regression or finer bins.
[2]	Ensemble of Deep Neural Networks for Estimating Particulate Matter from Images	Beijing PM Image Dataset — 1,460 outdoor photos ($\geq 584 \times 389$ px) with sky + depth; labels from U.S. Embassy PM _{2.5} data (no public release). Dataset: https://github.com/implicitDeclaration/HVAQ-dataset	Base: VGG-16, Inception-v3, ResNet-50 (ImageNet TL, 1-unit MSE). Meta: FFNN (3 → 32 → 64 → 32 → 1). 3-fold CV; Adam (0.001/0.01); batch 128; ≤ 150 epochs; early stop 10; Augment (h-flip, $\pm 2.5^\circ$ rot, crop 5–30 px $\times$ 10); TTA.	RMSE / R²: VGG-16 60.44 / 0.527 • ResNet-50 53.63 / 0.627 • Linear stack 51.63 / 0.655 • FFNN 49.37 / 0.684 (best) p < 0.0001.	1. Non-linear stacking beats single CNNs and simple averages. 2. Reproducible train pipeline + TTA. 3. Demonstrates transfer learning for visual air-quality tasks.	1. Imbalanced (low-PM dominant) dataset → poor high-PM prediction. 2. Only last FC layer fine-tuned. 3. Dataset not public.	1. Collect more high-PM images. 2. Add weather features (humidity, wind). 3. Test more CNN architectures & multi-modal fusion.

[3]	Classification Prediction Model of Indoor PM _{2.5} Concentration Using CatBoost Algorithm	Office building IAQ sensors (Beijing): 5-min data (2022-01-18 → 2022-03-29). 14,570 samples after cleaning (PM _{2.5} , RH, CO ₂ , Temp, Light). Binary label >25 µg/m ³ .	CatBoost GBDT (ordered boosting). Inputs: Prev PM, RH, CO ₂ , Temp, Light. Data cleaning (forward fill, RFMICE, 3-σ rule). Split 70/15/15. Optuna + Hyperband HPO (lr, depth, λ). Time-series CV. Baselines: MLP, GBDT, LR, DT, KNN.	CatBoost AUC-ROC 0.949, AUC-PRC 0.928, F1 0.883. Top features: Prev PM > RH > CO ₂ > Temp > Light.	1) Strong accuracy (AUC 0.949) with interpretable SHAP. 2) Time-series CV respects chronology. 3) Practical 5-feature input—easy to deploy with common IAQ sensors. 4) Thorough data cleaning pipeline documented.	1) Single-site, short window (~10 weeks) may limit generalization. 2) Binary label (25 µg/m ³) ignores multi-threshold actions (e.g., 15/35 µg/m ³ daily targets). 3) No explicit external validation across buildings/seasons.	1. Extend to multi-class thresholds (15/25/35 µg/m ³). 2. Multi-site/seasonal validation. 3. Add outdoor PM, ventilation state, occupancy proxies. 4. Tie predictions to closed-loop HVAC control for energy-aware actuation.
[4]	VDMS: An Improved Vision Transformer-Based Model for PM2.5 Concentration Prediction	PM2.5 data: Hourly ground measurements from CNEMC AOD data Meteorology: Daily gridded variables (wind, temperature, humidity, pressure) from national weather Data. Auxiliary data https://www.cnemc.cn/	Uses a Vision Transformer (ViT) to extract spatial features from satellite images, SimCLR for self-supervised pretraining to improve feature robustness, a	With Meteorological Data: Sam-CV R ² = 0.93, RMSE = 4.05 µg/m ³ , MAE = 3.23 µg/m ³ Sta-CV R ² = 0.88, RMSE = 5.57 µg/m ³ , MAE = 4.21 µg/m ³ Without Meteo	1. High Accuracy & Robustness 2. Strong Generalization 3. Multi-Source Data Fusion 4. Transformer-Based Self-Attention 5. Domain Knowledge	1. Limited Extreme-Event Representation 2. Data Smoothing Effects 3. Incomplete Spatiotemporal Modeling : The model treats spatial	1. Enhanced Data Quality & Coverage: Incorporate higher-resolution satellite data, expand geographic scope, and include more extreme pollution events to improve model

			Double-Layered LSTM (DLSTM) to capture temporal patterns, and a multi-head self-attention mechanism to fuse features and model global dependencies. The integrated design enhances spatial-temporal learning and prediction accuracy, even with sparse data.	Data: R^2 drops to 0.69–0.73. Baselines: CNN (0.75 R^2), ViT (0.53), SimCLR (0.86). Ablation: ViT-LSTM 0.58 → ViT-DLSTM 0.73 → VDM (+ Attention) 0.85 → VDMS (+ SimCLR) 0.93.	Integration	patterns as relatively static, missing dynamic interactions where pollution hotspots shift with time or weather changes. 4.High Computational Cost	robustness. 2.Advanced Spatiotemporal Modeling: Develop methods to better capture dynamic, time-evolving spatial pollution patterns and interactions between space and time. 3.Improved Representation of Heterogeneity: Account
[5]	PM2.5 concentration estimation based on image processing & linear regression.	Image sets used in Sensors study (photo-based) — image features extracted. Classic approach: compute image-level handcrafted features (e.g., saturation, contrast, haze-related indicators) and	Feature engineering (color/haze metrics) + simple linear regression.	RMSE = 11.88, R^2 = 0.41 ($p < 0.0001$). With exclusion ($RH \geq 65\%$ or $PM_{2.5} < 5$ removed): Region 1 improves to RMSE = 8.67, R^2 = 0.73 ($p <$	Low complexity; interpretable ; needs small data.	Performance sensitive to lighting/s cene; handcrafted features miss complex cues.	1.Combine with learned features 2. test on diverse field images 3.add meteorology

		map them to PM2.5 with simple regression		0.0001). The whole-image case is the worst; RoI selection consistently helps.			
[6]	Detection and prediction of air pollution using ML models	Beijing PM2.5 (UCI) Simple baseline classification and regression on PM2.5	Logistic Regression for pollution detection (0 / 1) and Autoregression (AR) for 7-day PM _{2.5} forecasting. Models trained/tested; accuracy and MSE evaluated.	Logistic Regression mean accuracy 0.998859 (std 0.000612) on their split. Prediction: AR 7-day-ahead MSE = 27.00; authors note MSE drops for shorter horizons	Very high accuracy for classification baseline; simple models	No DL; low generalization; no preprocessing	Extend to deep learning frameworks for temporal PM2.5 data
[7]	A Mobile Image-Driven PM2.5 Estimation Framework Using Deep Learning Techniques	CMU-HDR and DTU Datasets HDR: 442 image sequences (Chiang Mai, Thailand)	EfficientNet-B1 + BiLSTM model combining visual and temporal features, enhanced with	Best model: Meta-EfficientNet-B1-BiLSTM with 5.75% lower MAE, 11.43% lower SMAPE,	Combines image and weather data for accurate, efficient, and robust PM2.5 prediction	Limited to two regions and costly sequential data training; suffers from	Extend dataset to more diverse environment. Apply real-time deployment on mobile platforms.

	es	Non-HDR: 442 image sequences (Delhi, India) Includes meteorological data (temperature, humidity, wind speed)	meteorological data and evaluated using MAE, SMAPE, and Pearson's r.	and higher correlation; HDR images further improved accuracy.	n across real-world environments.	HDR/Non-HDR imbalance and sensitivity to lighting/weather changes.	
[8]	Enhancing PM2.5 Detection via Multispectral Image-Based Double-Channel Ensemble Learning for Drones	A double-channel ensemble learning model is proposed for PM2.5 detection, utilizing 23,100 multispectral and RGB images.	Dual-CNN, attention, GWAP, ensemble	DCEL-PM2.5 (proposed): 99.60% DCWCN: 93.76% VN8 (ViT-Net, 8 heads): 90.18% VGG16: 89.44%	High accuracy, low error, and stable across weather. Lightweight and real-time for UAVs. Effective feature fusion with attention	1. Green channel noise 2. Hardware cost (multispectral UAV) 3. Complex training architecture	Multi-location data expansion, Altitude and environmental variability, Lightweight model optimization, Real-world deployment
[9]	MMAQ: A Multi-Modal Self-Supervised Approach for	dataset : ~3 000 samples with Sentinel-2 (multi-spectral), Sentinel-5P (tropospheric	Multi-modal self-supervised model using Sentinel-2, Sentinel-5P, and	Linear regression: $R^2 = 0.586$ ($\uparrow$ from 0.50 baseline).	1. Captures inter & intra modal relations. 2. Requires no	1. High computational cost (3	1. Integrate meteorological and temporal

	Estimating Air Quality from Remote Sensing Data	columns), and tabular land-use data (altitude, population, area/station type). Also used fuel classification and plume segmentation dataset	tabular data. Based on Barlow Twins, extended for intra/inter-modal redundancy reduction with adaptive loss weighting. Uses ResNet-50 encoders for images and 3-layer MLP + attention for tabular input. Augmentations: affine, flip, band-drop, masking, scaling	Fine-tuning: $R^2 = 0.751$ (best, MSE = 32.6, MAE = 4.51) $\rightarrow \approx 17\%$ improvement over AQNet. Transfer tasks: Cls 95.9 %, Seg mIoU 75.5.	labels for pretraining. 3. Strong generalization across tasks. 4. Adaptive loss balancing auto-tunes learning.	encoders + attention) . 2. Complex loss tuning and augmentation design. 3. Dataset small (~3 K samples).	data. 2. Extend to other pollutants & countries. 3. Improve scalability for larger datasets and real-time forecasting.
[10]	Self-supervised air quality estimation with graph neural network assistance and attention enhancement	35 stations, hourly $PM_{2.5}/PM_{10}/NO_2/CO/SO_2/O_3$ + meteo. UK (2016–2019) : 141 stations (reduced to 30 by ≤ 200 km rule), daily $PM_{2.5}/PM_{10}/NO_2/O_3/SO_2$ + meteo. Station-level splits	Encoder: GCN+GRU T-GCN trained with adversarial contrastive MI (graph corruptor + discriminator). Wind-aware node features and distance-weighted interpolation. Decoder: dual attention (location & feature) + FC using target meteo + wind score. MSE loss.	MAE 10.6, RMSE 14.12, R^2 0.88; UK MAE 2.16, RMSE 3.19, R^2 0.59. Improves MAE by 4.93–34.88 % (UK) and 6.89–31.94 % (Beijing) vs baselines.	1. Captures spatial+temporal structure; 2. self-supervised encoder; 3. wind-aware & attention boosts; 4. code released.	1. Depends on neighbor trends; 2. assumes ≤ 200 km neighbor availability; 3. spatial contributes slightly more than temporal—may underperform with very sparse	1. Explore robustness with sparser nets 2. alternate graph priors beyond inverse distance 3. richer wind/meteorology sources 4. uncertainty quantification

						networks.	
[11]	Semi-Supervised Air Quality Forecasting via Self-Supervised Hierarchical Graph Neural Network	Two real-world datasets — BTH (Beijing–Tianjin–Hebei) and PRD (Pearl River Delta). Each includes hourly pollutants (AQI, PM _{2.5} , PM ₁₀ , O ₃ , NO ₂ , SO ₂ , CO), meteorology (temperature, humidity, pressure, wind), forecast weather, and urban context (POI, road network, traffic). Regions with stations split 3-fold (train 2 / test 1). Metrics: MAE, RMSE.	Semi-supervised hierarchical GNN combining: (1) Multi-view AQ estimation (spatial KNN + temporal GRU + context POI/road), (2) Hierarchical GNN (city → zone → region) with gated GRU, and (3) Self-supervised neighbor + context reconstruction. Loss = MSE + α Aux + β Self.	BTH: MAE ↓ 15.9 / 14.6 / 12.6 %, RMSE ↓ 14.9 / 10.5 / 10.2 % for 1–6 h / 7–12 h / 13–24 h horizons. PRD: MAE ↓ 13.8 / 16.1 / 18.4 %, RMSE ↓ 13.6 / 17.1 / 13.9 %. Outperforms DeepAir, GeoMAN, GC-DCRN N, MGED-Net, MasterGNN, and SHARE on all horizons.	1. Captures multi-scale spatial & temporal dependencies (city–zone–region). 2. Handles unmonitored regions using semi- and self-supervision. 3. Incorporates forecast weather and urban context for robust prediction. 4. Achieves state-of-the-art accuracy on both datasets.	1. Complex model with many hyperparameters (α , β , K, nz, T). 2. Requires rich auxiliary data (POI, traffic, roads). 3. Slight performance drop on extreme AQI cases.	Extend to other regions and pollutants, integrate uncertainty estimation, improve real-time forecasting efficiency, and enhance interpretability for policy applications.

References:

- [1] <https://ieeexplore.ieee.org/abstract/document/8297023>
- [2] <https://ieeexplore.ieee.org/abstract/document/8492790>
- [3] <https://www.frontiersin.org/journals/built-environment/articles/10.3389/fbuil.2023.1207193/full>
- [4] <https://www.mdpi.com/2076-3417/15/13/7346>
- [5] <https://www.sciencedirect.com/science/article/pii/S0360544223034394>
- [6] https://www.researchgate.net/profile/Aditya-C-R/publication/327746379_Detection_and_Prediction_of_Air_Pollution_using_Machine_Learning_Models/links/5f12668ea6fdcc3ed7120693/Detection-and-Prediction-of-Air-Pollution-using-Machine-Learning-Models.pdf
- [7] <https://ieeexplore.ieee.org/abstract/document/10813353>
- [8] <https://ieeexplore.ieee.org/abstract/document/11098626>
- [9] <https://ieeexplore.ieee.org/document/10647792>
- [10] <https://link.springer.com/article/10.1007/s00521-024-09637-7>
- [11] <https://ieeexplore.ieee.org/document/9709128>